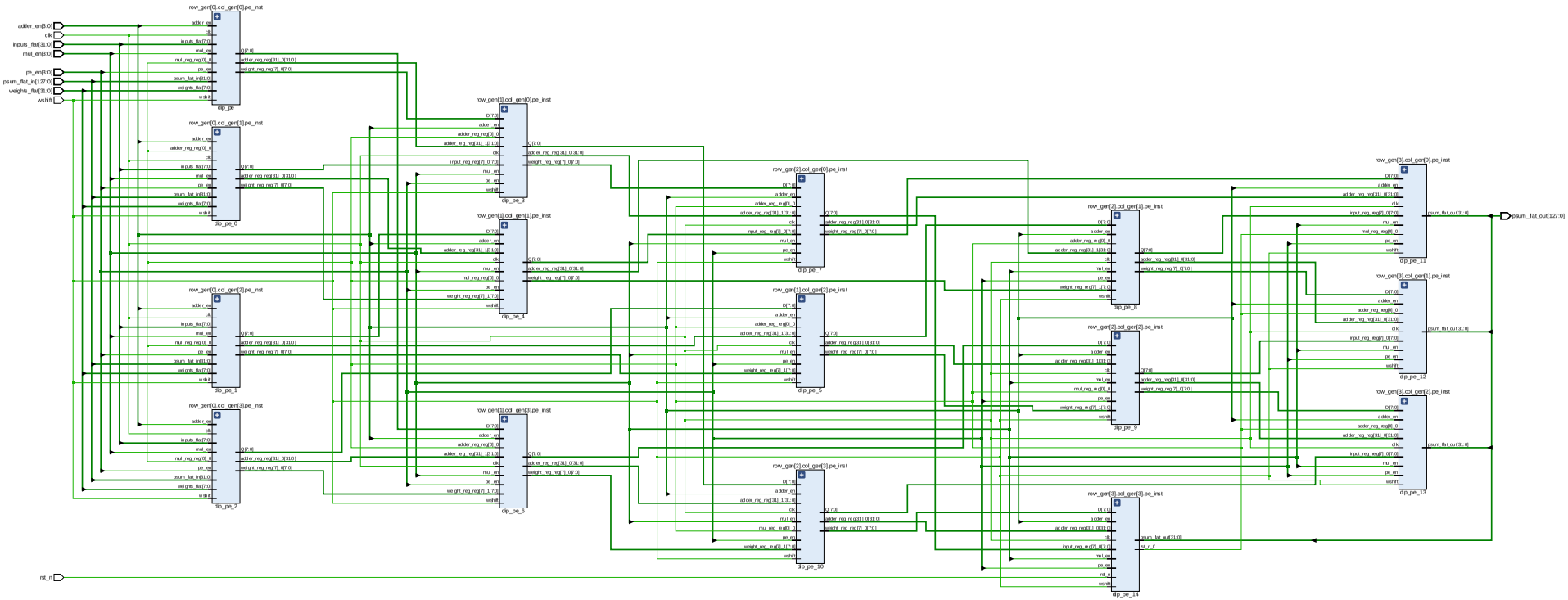

adder_en[3:0]
clk
inputs_flat[31:0]
mul_en[3:0]
pe_en[3:0]
psum_flat_in[127:0]
weights_flat[31:0]
wshift
rst_n
psum_flat_out[127:0]
row_gen[0].col_gen[0].pe_inst
dlp_pe
row_gen[0].col_gen[1].pe_inst
dlp_pe_0
row_gen[0].col_gen[2].pe_inst
dlp_pe_1
row_gen[0].col_gen[3].pe_inst
dlp_pe_2
row_gen[1].col_gen[0].pe_inst
dlp_pe_3
row_gen[1].col_gen[1].pe_inst
dlp_pe_4
row_gen[1].col_gen[2].pe_inst
dlp_pe_5
row_gen[1].col_gen[3].pe_inst
dlp_pe_6
row_gen[2].col_gen[0].pe_inst
dlp_pe_7
row_gen[2].col_gen[1].pe_inst
dlp_pe_8
row_gen[2].col_gen[2].pe_inst
dlp_pe_9
row_gen[2].col_gen[3].pe_inst
dlp_pe_10
row_gen[3].col_gen[0].pe_inst
dlp_pe_11
row_gen[3].col_gen[1].pe_inst
dlp_pe_12
row_gen[3].col_gen[2].pe_inst
dlp_pe_13
row_gen[3].col_gen[3].pe_inst
dlp_pe_14